

Phase 1: The Basics (Foundations of Inference)

1. Probability Distributions

A **Probability Distribution** is a mathematical function that provides the probabilities of occurrence of different possible outcomes for an experiment.

Key Concepts

Random Variable: A variable whose values depend on outcomes of a random phenomenon.

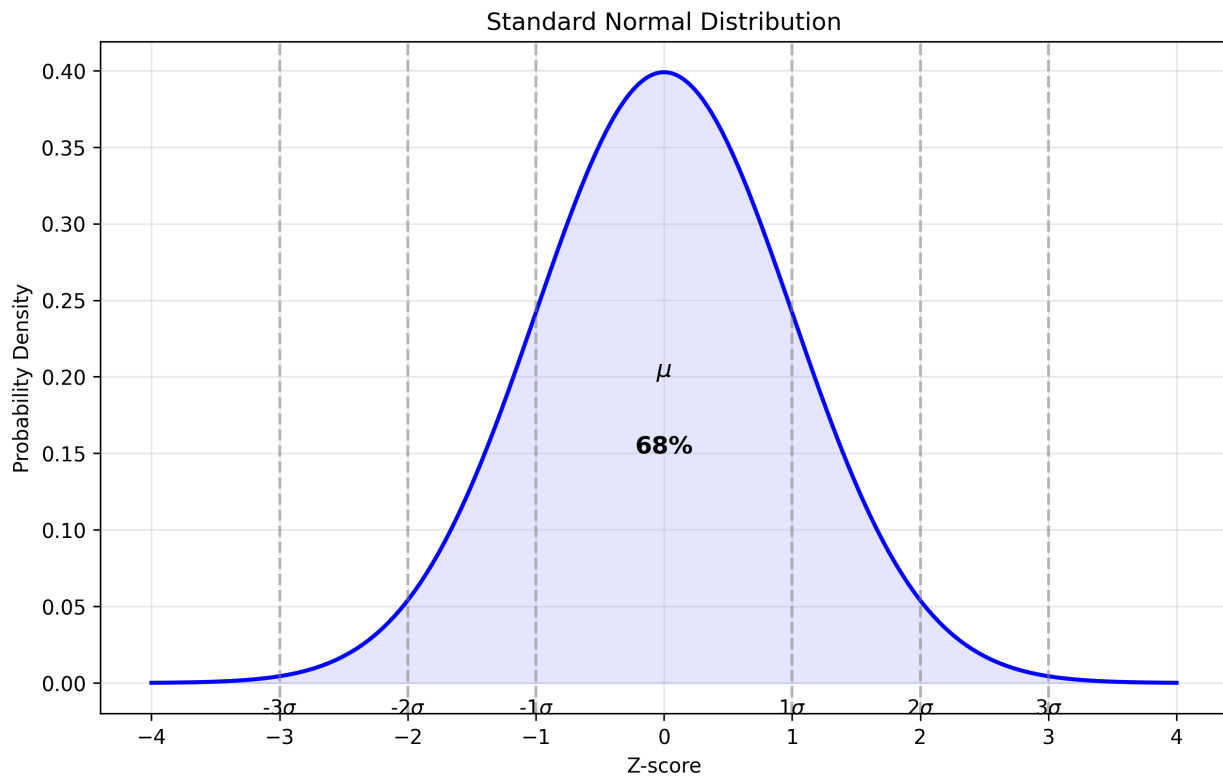
- **Discrete:** Countable outcomes (e.g., number of heads in 10 coin flips).
- **Continuous:** Infinite possible values within a range (e.g., height, time).
- **PMF (Probability Mass Function):** Used for **Discrete** variables. Gives the probability that a discrete random variable is exactly equal to some value.
- **PDF (Probability Density Function):** Used for **Continuous** variables. The probability of the variable falling within a particular range of values is given by the integral of the variable's density over that range. **Note:** The probability of a continuous random variable taking exactly any single value is 0.
- **CDF (Cumulative Distribution Function):** The probability that the variable takes a value less than or equal to x . Works for both discrete and continuous.

Common Distributions in Data Science

Distribution	Type	Description	Interview Use Case
--------------	------	-------------	--------------------

Normal (Gaussian)	Continuous	Bell-shaped, symmetric. Defined by Mean (μ) and Std Dev (σ). 68-95-99.7 Rule.	Modeling natural phenomena (height, errors). Assumption for many tests (t-test, Z-test).
Bernoulli	Discrete	Single trial with 2 outcomes (Success/Failure). p = prob of success.	Modeling a single user click (Click/No Click).
Binomial	Discrete	Number of successes in n independent Bernoulli trials.	Predicting number of users who convert out of 1000 visitors.
Poisson	Discrete	Number of events in a fixed interval of time/space. Constant mean rate λ .	Modeling arrival of tickets in a support queue or server requests per minute.
Exponential	Continuous	Time <i>between</i> events in a Poisson process. Memoryless property.	Predicting time until the next customer purchase or server failure.
Uniform	Continuous	All values in range $[a, b]$ are equally likely.	Random number generation; null model where no outcome is preferred.

Visual: Standard Normal Distribution



2. The Central Limit Theorem (CLT)

The Holy Grail of Statistics.

Definition

The CLT states that the **sampling distribution of the sample mean** will be approximately **normally distributed**, regardless of the population's distribution, provided the sample size is sufficiently large (usually $n \geq 30$).

Why it matters for Data Science?

- **Inference on Non-Normal Data:** Even if your user data is skewed (e.g., LTV or session duration is exponential/log-normal), if you take the mean of enough users, that mean follows a normal distribution.

- **A/B Testing:** This allows us to use parametric tests (like t-tests) on metrics like Average Revenue Per User (ARPU) even if the underlying revenue data is non-normal.

Law of Large Numbers (LLN)

As the sample size (n) grows, the sample mean ($\bar{x}$) gets closer and closer to the true population mean (μ).

* **Difference from CLT:** LLN tells you *where* the center is going (accuracy); CLT tells you about the *shape* of the distribution around that center (precision/normality).

3. Hypothesis Testing Fundamentals

The formal procedure to accept or reject statistical statements.

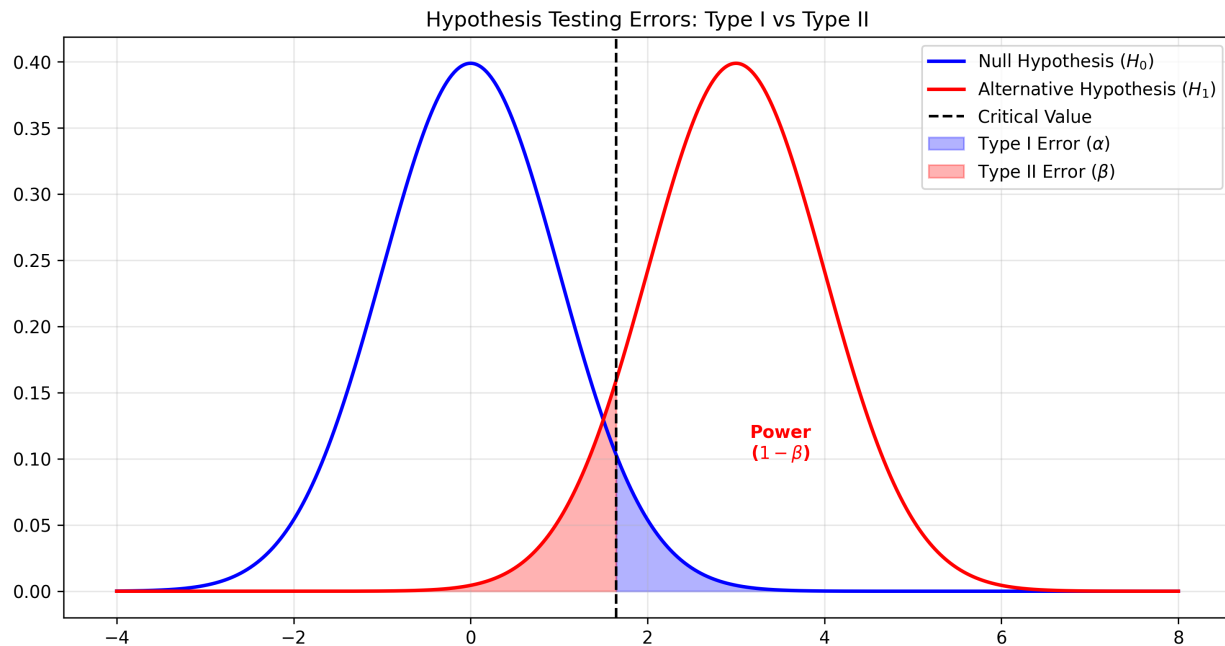
Core Terminology

- **Null Hypothesis (H_0):** The default state. "There is no difference," "The coin is fair," "The new feature changed nothing."
- **Alternative Hypothesis (H_1 or H_a):** The claim you want to test. "There is a difference," "The coin is biased."
- **Significance Level (α):** The probability of rejecting H_0 when it is actually true. Usually set to 0.05 (5%).
- **Power ($1 - \beta$):** The probability of correctly rejecting H_0 when it is false. Usually aim for 0.80 (80%).

P-value: The probability of observing a test statistic as extreme as, or more extreme than, the one observed, **assuming H_0 is true**.

- *Interview Tip:* Never say "Probability that H_0 is true." Say "Probability of data given H_0 ."
- **If $p < \alpha$:** Reject H_0 (Statistically Significant).
- **If $p \geq \alpha$:** Fail to reject H_0 .

Visual: Hypothesis Testing Errors



Errors Matrix

Decision	H_0 is True	H_0 is False
Reject H_0	Type I Error (α) "False Positive"	Correct Decision (Power)
Fail to Reject H_0	Correct Decision	Type II Error (β) "False Negative"

4. Standard Error (SE) vs Standard Deviation (SD)

The most common "gotcha" question.

Standard Deviation (SD)

- **Measures:** Variability of the **data points** themselves.
- **Context:** Descriptive statistics. "How spread out are the heights of people in this room?"

- **Formula:**

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

Standard Error (SE)

- **Measures:** Variability of the **sample mean** (or other statistic).
- **Context:** Inferential statistics. "If I took 100 different samples of people, how much would the average height vary across those samples?"
- **Formula:**

$$SE = \frac{\sigma}{\sqrt{n}}$$

- **Key Insight:** SE decreases as sample size (n) increases (because dividing by sqrt(n)). SD does *not* decrease with n (it stabilizes).

5. Z-test vs T-test (Parametric Tests)

Formulas

- **Z-Score:**

$$Z = \frac{\bar{x} - \mu}{SE}$$

- **T-Score:**

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Comparison Table

Feature	Z-test	T-test
Population Variance (σ^2)	Known	Unknown (Estimated by sample s^2)
Sample Size (n)	Generally Large ($n > 30$)	Small ($n < 30$) (though often used for large n too via software)
Distribution	Standard Normal Z-distribution	Student's t-distribution (Heavier tails than Normal)

Degrees of Freedom	N/A	df = n - 1 (flatter tails for lower df)
---------------------------	-----	---

Interview Rule of Thumb: In the real world, we rarely know σ , so we almost always use **T-tests** (or Welch's T-test for unequal variances), even for large samples, because the T-distribution converges to Normal as $n \rightarrow \infty$.

References

- [Verve Copilot - Probability Distributions](#)
- [GeeksForGeeks - Central Limit Theorem](#)
- [Medium - Hypothesis Testing Basics](#)
- [Career Foundry - SE vs SD](#)